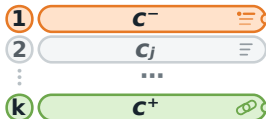


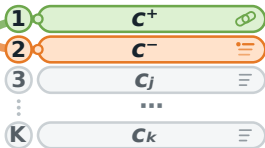
Initial retrieval for q_i
top-K

$$\Delta_i < 0$$



Refined ranking

$$\Delta_i > 0$$



Failure-Calibrated Refinement

$$\Delta_i = s(q_i, c^+) - s(q_i, c^-)$$

$$< 0$$

$$> 0$$



Paired code



Failure Candidate



Other candidate